

Chase M. Enlowsmith

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EDUCATION	The University of Texas at Austin , Austin, Texas Dec 2025 Bachelor of Science in Physics & Astronomy • GPA 3.91 • Relevant Coursework: Quantum Mechanics, Intro to Quantum Field Theory, Electrodynamics, Thermodynamics, Relativity, Astrophysics, Cosmology
HONORS AND AWARDS	Phi Beta Kappa , Alpha of Texas chapter Feb 2025 – Current University Honors , University of Texas at Austin May 2023 – Dec 2025 Lightning Talk Competition - 2nd Place , Applied Research Labs Jul 2025 2025 Honors Scholar Lightning Talk: <i>“Machine Learning on Sound Speed Profiles to Characterize Ocean Variability”</i>
PROFESSIONAL RESEARCH EXPERIENCE	Applied Research Lab , Austin, Texas Jun 2025 – Jun 2026 Honors Scholar Intern • Engineered a machine learning pipeline to geographically cluster underwater acoustic data across basin scales. • Characterized ocean variability on decadal timescales, providing valuable insight into long-term ocean patterns. • Created a bespoke data-reduction methodology to optimize pipeline accuracy and efficiency beyond industry standards. • Innovated new machine learning approaches to predict acoustic features for tactical oceanography applications. • Authoring a manuscript to be published in peer-reviewed journals. Center for Gravitational Physics , U. Texas, Austin Oct 2024 – Aug 2025 Undergraduate Researcher • Designed and developed a data analysis pipeline to calculate galaxy redshifts from the Hobby-Eberly Telescope (HET) VIRUS-P Parallel Survey (VIPS) utilizing the Redrock API. • Provided the extracted redshifts from HET spectra for current comparative studies with gravitational wave-derived distance measurements to further investigate the Hubble tension. • Investigated how cosmological assumptions, the number of lines of sight, galaxy catalogue sparseness, and EM-signal availability all affect the Hubble constant posterior.
RESEARCH PROJECTS	CMBverse , U. Texas, Austin Jun 2024 – Present • Built an educational website with interactive graphical tools for academics to explore CMB power spectra across a vast range of Lambda Cold Dark Matter (LCDM) cosmologies. The most versatile and dynamic teaching tool for CMB studies. • Developed an end-to-end Einstein-Boltzman Solver emulator able to speed up power spectra calculations by over 1000x while maintaining sub-percent level accuracy across relevant cosmologies, powered by the Cosmological Neural Network Emulator of CLASS using TensorFlow (CONNECT). • Authoring a manuscript to be published in peer-reviewed journals.

	Galaxy Cluster Mass Bias , U. Texas, Austin Jun 2024 – Present • Built random-forest and symbolic-regression pipelines to derive closed-form analytic proxies for galaxy-cluster mass and hydrostatic mass bias from observable X-ray and SZ features, using MillenniumTNG cosmological simulations. • Reproduced and validated an analytic correction to the Sunyaev–Zel’dovich flux–mass relation, reducing cross-validated prediction scatter by 20–30%. • Designing an autonomous LLM-agentic discovery workflow with a leakage-resistant scoring harness to predict galaxy cluster mass, measuring susceptibility of ML-driven science to various implementation schemes and automated integrity auditing. • Authoring a manuscript to be published in peer-reviewed journals.
	Gravitational Lens Detection , U. Texas, Austin Sep 2025 – Present • Developing a machine learning pipeline to accurately identify gravitationally-lensed galaxies within a sample of luminous red galaxies in the Kilo Degree Sky Survey data release 5. • Constructing a robust data analysis tool that first maps galaxy images into a statistical latent space via a variational autoencoder, and secondly applies a t-distributed stochastic neighbor embedding to capture new galaxy-lens candidates.
	GW Data Sonification , U. Texas, Austin Sep 2025 – Dec 2025 • Constructed a command-line tool to download, visualize, and sonify gravitational wave events from GWOSC (Gravitational Wave Open Science Center). • Developed an open-source pipeline to extract black hole merger parameter distributions from LIGO, construct waveforms, and map into an audio format. • Built a Jupyter Notebook and detailed guide detailing an extend explanation of pipeline usage, the theoretical physics of gravitational waves, and gravitational wave observation.
LEADERSHIP AND COMMUNITY INVOLVEMENT	Theta Chi Fraternity , U. Texas, Austin Scholarship Chair Dec 2023 – Dec 2024 • Connected 40+ members with applicable scholarship opportunities, career events, and academic opportunities. • Oversaw academic performance and provided one-on-one career and academic assistance. Recruitment Captain Dec 2023 – Dec 2024 • Coordinated recruitment events and outreach to potential new members.
	UT MicroFarm , Austin, Texas Volunteer Apr 2023 – May 2024 • Worked with a team of volunteers to grow plants, manage the upkeep of community garden plots, and compost. • Contributed to producing dozens of pounds of fresh produce donated to the student food bank.
TECHNICAL SKILLS	Python, Matlab, Unix, L ^A T _E X, Git, Microsoft Office Suite, Google Suite, Data Analysis, Machine Learning
LANGUAGES	English (Native), Spanish (Basic)